Unsupervised Domain-Invariant Feature Learning for Cloud Detection of Remote Sensing Images

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***Abstract*— The detection of clouds in remote sensing (RS) images is an important task, and convolutional neural networks (CNNs) have been used to perform it. However, supervised cloud detection CNNs rely heavily on a large number of samples annotated at pixel level to tune their parameter. Annotating RS images is a labor-intensive procedure and requires expert-level human knowledge. To reduce the labeling cost, we propose an unsupervised domain adaptation (UDA) approach to enable the model trained on labeled source satellite images to generalize to unlabeled target satellite images. Specifically, we propose a fine-grained feature alignment (FGFA) domain adaptation strategy that encourages a cloud detection network to extract domain-invariant representations, which improves the accuracy of cloud detection in unlabeled target satellite images. The proposed FGFA strategy consists of two steps: 1) fine-grained class-relevant feature selection based on an attention-guided mechanism and 2) class-relevant feature alignment (FA) based on a proposed grouped FA approach. Experimental results on the "Landsat-8 → ZY-3" and "GF-1→ ZY-3" domain adaptation tasks demonstrate the effectiveness of our method and its superiority to existing state-of-the-art UDA approaches.**

***Index Terms*— Cloud detection, domain-invariant feature, fine-grained feature alignment (FGFA), remote sensing (RS) imagery.**

I. Introduction

CLOUD detection has attracted increasing attention over the past decade. In particular, the adoption of open data policies by major Earth observation programs (e.g., NASA's Landsat and ESA's Sentinel) has promoted the development of various cloud detection algorithms [1]. The development of deep learning [2] has contributed to major breakthroughs in image processing tasks, such as semantic segmentation [3], image classification [4], image restoration [5], and target detection [6].

Manuscript received January 4, 2021; revised May 26, 2021, August 22, 2021, and October 4, 2021; accepted October 9, 2021. Date of publication October 14, 2021; date of current version February 3, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61771339, Grant 62171317, Grant 62171309, and Grant 62072331. *(Corresponding authors: Jingyu Yang; Huanjing Yue.)*

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Digital Object Identifier 10.1109/TGRS.2021.3120001

In recent years, advanced convolutional neural network (CNN) models have been proposed to detect clouds from multispectral satellite images. These networks include classical superpixel classification [7], [8], typical end-to-end CNN-based frameworks [1], [9], and multiscale/level feature fusion networks [10], [11]. However, such networks rely heavily on a large number of samples annotated at pixel level for parameter tuning, and sample collection is tedious, time-consuming, and expensive [12]. In our previous ZY-3 thumbnail cloud detection work [10], [13], we also developed supervised CNN frameworks and confronted the headache problem in labeling remote sensing (RS) imagery. There are ten people in the labeling group for our dataset. Specifically, before labeling, group members are trained by experts from the Land Satellite Remote Sensing Application Center (LSRSAC) of China. For each image, we obtain the pixel-level annotation by using an image editing software, such as Adobe Photoshop. Each labeled image is cross-checked by two other group members. Besides, tough cases with inconsistent labeling are further checked and refined by experts from LSRSAC. It usually takes 1–2 h to label pixelwise annotation from a cloud-covered thumbnail. Therefore, unsupervised learning approaches are highly demanded to reduce the human effort of manually labeling datasets.

Nowadays, there are available cloud cover evaluation datasets, such as the Landsat-8 Cloud Cover Assessment Validation Dataset [14] and GF-1 WFV Cloud and Cloud Shadow Cover Validation Dataset [15]. However, it is difficult to directly transfer CNN-based methods trained on these existing datasets to analyze target satellite images because of potentially quite different domain characteristics. This issue is known as the domain shift problem [16] due to radiation resolution, spatial resolution, and spectral range of sensors as well as high dynamics of cloud shape and type of land cover. For example, in Fig. 1, we show texture maps of satellite images obtained by different sensors. Results in Fig. 1 demonstrate that clouds in the obtained RS images of different satellite sensors show large texture style differences. Although cloud region is mainly bright and white (saturated) in almost all the satellite images, there is a large domain distribution gap between these different sensors obtained clouds. Therefore, in order to make full use of existing datasets, an effective domain alignment method is urgently needed to reduce the domain differences between different datasets.

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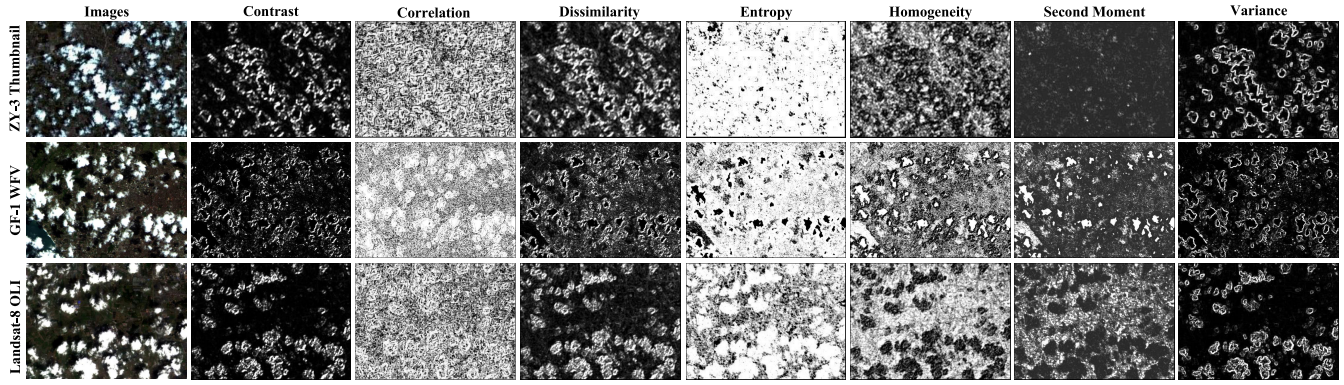

Fig. 1. Texture maps of satellite images obtained by different sensors. Texture maps include contrast map, correlation map, dissimilarity map, entropy map, homogeneity map, second moment map, and variance map. Since the radiation resolutions of ZY-3 thumbnail, GF-1 WFV, and Landsat-8 OLI are 8, 10, and 12 bit, respectively, we generate feature maps from normalized images (normalized to 0–1) for a fair comparison.

One effective way to address the domain shift problem is to utilize a domain adaptation strategy to force the feature extractor to generate domain-invariant features [16], [17]. With the development of generative adversarial network (GAN), unsupervised domain adaptation (UDA) approaches for image segmentation based on adversarial learning, such as adversarial discriminative domain adaptation (ADDA) method [18], cycle-consistent adversarial domain adaptation (CyCADA) method [19], and adversarial entropy (AdvEnt) minimization method [20], have attracted much attention. However, most UDA-based semantic segmentation approaches were developed for natural street-view images, such as those in the Cityscapes Dataset [21], which were acquired using a general camera. Optical satellite RS images are very different from those of natural street-view images. As noted above, in the task of cross-satellite image cloud detection, spectral and resolution differences exist in images acquired by different satellite sensors, which results in large domain discrepancies that make it difficult to generalize the source-trained model to the target dataset. In addition, the domain shift problem is caused by the high dynamics in cloud shape and different cloud categories as well as different global geomorphic types. Therefore, it is a great challenge to use a simple UDA method for cloud detection from cross-satellite images.

In this article, we focus on using the UDA method to obtain cloud coverage information from thumbnails of ZY-3 images to assess their quality. As illustrated in [10] and [13], thumbnails of high-resolution satellite images contain the information needed by data producers to obtain cloud coverage information. Nevertheless, a thumbnail image contains an RGB image (with three bands) or a gray image (with only one band). There is no doubt that there is a large domain difference between thumbnail images and hyperspectral or multispectral RS images. To reduce the domain discrepancy gap, we propose the use of feature-level domain adaptation to encourage a cloud detection network to generate domain-invariant features for source and target domain datasets. Inspired by class-level domain adaptation approach [22], we propose a fine-grained feature alignment (FGFA) domain adaptation method that considers not only local class-relevant feature domain shift problems between the source and target domain datasets but also the effective alignment of local class-relevant feature. To be specific, we propose a class-relevant feature selection mechanism to select a fine-grained class-relevant feature and then propose a feature alignment (FA) strategy to effectively align the selected class-relevant features of the source and target domain datasets into a common feature space.

To evaluate the effectiveness of the proposed method, we use two large-volume RS cloud detection datasets, the Landsat-8 Cloud Cover Assessment Validation Dataset [14] and GF-1 WFV Cloud and Cloud Shadow Cover Validation Dataset [15], to form the labeled source domain dataset, whereas the unlabeled ZY-3 satellite thumbnail dataset is used as the target domain. In this article, the final goal is to learn a cloud detection model that is trained from a source-labeled domain satellite dataset and can perform well on the target ZY-3 dataset. Our contributions can be summarized as follows.

1) We propose an FGFA strategy to encourage a segmentation network to extract domain-invariant representations. The aim is to improve cloud detection accuracy on target unlabeled satellite images, and the effectiveness of the strategy is demonstrated for two domain transfer tasks.
2) We propose an attention-guided (AG) class-relevant feature selection mechanism in which spatial attention maps obtained from the predicted score maps of the segmentation network are used to guide the selection of fine-grained class-relevant features in the source and target domain datasets.
3) We propose a class-relevant FA method based on the grouped feature alignment (GFA) strategy that is able to effectively reduce feature distribution gaps between the source and target domains to obtain domain-invariant feature representations.

This article is organized as follows. In Section II, we present the related work of UDA approaches for image segmentation. In Section III, we present the proposed UDA framework in detail, including AG class-relevant feature selection mechanism, GFA strategy, and network optimization learning strategy. The experimental settings and dataset are presented in

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Section IV. Experimental results and discussions are presented in Section V, followed by conclusions in Section VI.

II. Related Work

A. CNN-Based Cloud Detection

In recent years, with development of deep learning technology, deep CNNs have achieved state-of-the-art performance in image classification, segmentation, and object detection [2]. Deep learning technology has also contributed to a major breakthrough in the RS image cloud detection task. Many CNN-based cloud detection methods have been proposed. In this article, we roughly classified these methods into the following three categories.

1) Supervised Superpixel Classification Methods: Inspired by image classification based on fully connected deep CNNs (such as AlexNet [23], GoogLeNet [24], and ResNet [25]), Xie *et al.* [7], Morales *et al.* [8], and Matthieu-Le *et al.* [26] directly brought image classification methods to detect cloud from RS images. They have used the simple linear iterative cluster (SLIC) [27] to obtain the superpixel and then predicted a category for each superpixel by using the classification networks. However, the performance of these methods is limited by the accuracy of superpixel clustering. To further optimize the segmentation results, postprocessing optimization strategy has been introduced to refine cloud detection results. For example, Yue *et al.* [28] used a fully connected conditional random field (CRF) model [29] to refine the cloud detection results to obtain accurate cloud boundaries. However, postprocessing optimization is time-consuming for large size of RS image. Postprocessing optimization cannot achieve end-to-end training and testing.

2) Supervised Semantic Segmentation Methods: Fully convolutional neural networks (FCNNs) have achieved remarkable results in semantic segmentation of traditional camera images, such as PASCAL VOC2012 [30]. Due to the advanced semantic segmentation performance of FCNNs, RS image cloud detection has also achieved remarkable results. Classical FCNN methods, such as U-net [31] and SegNet [32], have been introduced to detect cloud from satellite images. For example, Jeppesen *et al.* [1], Mohajerani *et al.* [33], Drönner *et al.* [34], and Wieland *et al.* [35] proposed a cloud detection model based on the modified U-net framework. Zhang *et al.* [36] and Ghassemi and Magli [37] developed lightweight networks based on U-net variants for onboard RS cloud detection. Also, Chai *et al.* [9] and Lopez *et al.* [38] developed a multispectral RS image cloud detection model based on a modified SegNet [32]. Other advanced cloud detection network structures, such as multiscale/multilevel feature fusion networks [10], [13], [39], were also proposed. Besides, strategy-based advanced methods, such as deep matting-based method [40] and geographic information-driven method [41], are proposed to detect clouds. Overall, the above-mentioned CNN-based cloud detection networks are able to achieve promising performance when training with a large number of pixel-level annotated samples. However, labeling pixel-level annotations is tedious, time-consuming, and expensive, which is not conducive to the generalization of supervised learning models.

3) Weakly Supervised/Unsupervised Semantic Segmentation Methods: Most of the existing CNN-based cloud detection methods are based on the supervised framework that heavily relies on a large number of training data with strong pixelwise annotations. Therefore, several recent research efforts have focused on detection clouds based on weakly supervised/unsupervised methods to reduce the human effort of manually labeling datasets. Drawing on the idea of weakly supervised semantic segmentation [42], Yansheng *et al.* [43] proposed a novel framework that only relies on image-level class labels for pixel-level cloud detection of RS image. In recent years, with the development of GAN technology [44], cloud detection based on the GAN framework has attracted much attention. For example, Guo *et al.* [45] proposed a weakly supervised cloud detection network developed from a GAN-based domain translation network, which uses cloud and nonclouds images as model training data. Xie *et al.* [46] proposed an unsupervised cloud detection network for multispectral and hyperspectral RS images based on adversarial learning technology. Overall, cloud detection based on weakly supervised/unsupervised semantic segmentation approaches is still in the initial stage, and many problems remain to be solved.

B. Unsupervised Domain Adaptation

Adversarial training for UDA is the most explored approach for image semantic segmentation [20]. A standard adversarial learning-based UDA framework usually contains two networks, i.e., a semantic segmentation network and a domain discriminative network. The domain discriminative network functions as a domain discriminator that supervises the semantic segmentation model so that it learns domain-invariant features from different domain datasets. This encourages the network trained on the source domain dataset to be well generalized to the target domain dataset [47]. In the past few years, UDA approaches to image semantic segmentation have mainly focused on reducing the domain distribution gaps using input-level (pixel-level) [48], feature-level [49], and output-level [20] domain adaptations.

For cross-satellite RS images domain adaptation, simple visual appearance-based domain adaptation methods, such as style transfer approaches [50], have been introduced to reduce domain distribution gaps [51]–[53]. These methods achieve visual appearance-based domain adaptation by generating a fake target domain that has a visual appearance similar to that of the target domain data. However, these methods only focus on the visual appearance aspect of domain adaptation and ignore the semantic alignment, which would effectively reduce domain distribution gaps. If domain adaptation focuses on visual appearance alone for RS images, it may not perform well because of the complex and diverse subdomains in the source and target satellite datasets. In recent years, UDA semantic segmentation approaches based on feature-level domain adaptation, such as DAAD [18], CyCADA [19], and deep activation matching (DAM) [54], have attracted much interest. However, these methods focus on aligning the marginal distribution in the feature space and ignore the shift in the

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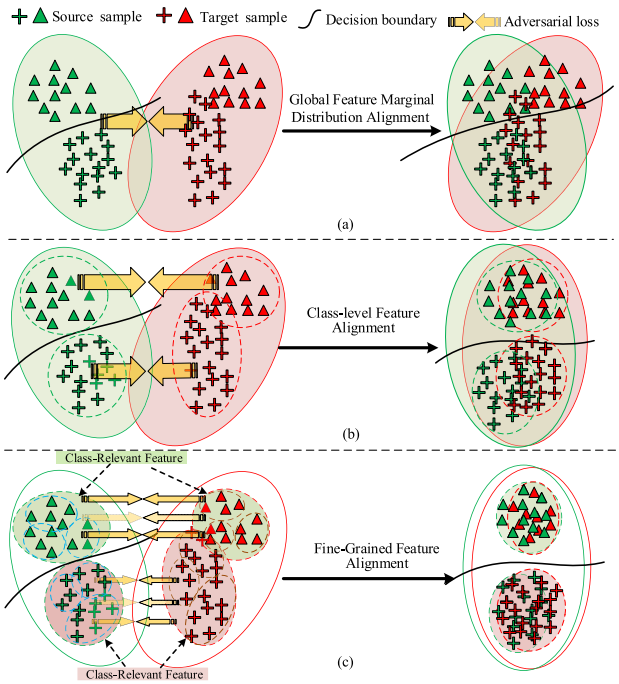

Fig. 2. Illustration of traditional, class-level, and fine-grained class-relevant FA. (a) Traditional global feature marginal distribution alignment. (b) Class-level FA, where features are expected to be aligned according to specific classes. (c) Proposed FGFA focuses on aligning each class-relevant feature with the proposed GFA strategy.

local joint distribution [55]. In other words, feature marginal distribution alignment strategy does not consider local categorywise feature distribution shifts, which inevitably results in under transfer or negative transfer, as shown in Fig. 2(a).

Recently, FA domain adaptation based on categorywise [55] and class-level [22] approaches has been proposed to adaptively align each class and reduce the local domain shift. As shown in Fig. 2(b), although these methods focus on the local domain shift, under transfer problem still exists. Unlike methods in [22] and [55], we consider how to suppress undesirable class-irrelevant features to promote positive transfer and combat negative transfer. In this article, we propose an AG class-relevant feature selection mechanism that can suppress undesirable class-irrelevant features and activate the class-relevant features of the source and target domain datasets. Unlike traditional holistic feature alignment (HFA) strategy used in [18], [19], [22], [54], and [55], we propose a GFA strategy to effectively align the class-relevant features of the source and target domain datasets into a common feature space. Our method first splits the class-relevant feature into different groups and then applies domain adaptation via an adversarial learning approach. As a result, it is possible to align each feature group in the source and the target data independently. Fig. 2(c) shows the proposed domain adaptation operation.

III. Proposed Method

A. Proposed UDA Framework

To effectively utilize the existing labeled source dataset and address the domain shift problem, we employ a UDA strategy to reduce the domain distribution gaps between the source and target satellite images. In Fig. 3, we present our proposed UDA framework, which is based on the FGFA strategy, for detection clouds in RS images. To achieve FGFA domain adaptation, we first propose an AG class-relevant feature selection method that can help accurately select the semantic feature maps of each class and then align these feature maps into a common feature space using the proposed GFA strategy. Specifically, the proposed FGFA domain adaptation consists of two stages: AG class-relevant feature selection and class-relevant FA.

Stage I (AG Class-Relevant Feature Selection): Before local class-relevant feature is aligned, the undesirable effects of class-irrelevant feature should be suppressed. To do this, feature selection is used. Specifically, we use spatial attention maps obtained from the predicted score maps of the segmentation network to guide the selection of fine-grained class-relevant feature.

Stage II (Class-Relevant FA): To align class-relevant feature, we propose a GFA strategy that helps align features in source and target domain datasets into a common feature space. In this way, we can encourage a segmentation network to generate domain-invariant features.

B. AG Class-Relevant Feature Selection

Inspired by [56], we use spatial attention maps obtained from the predicted score maps of the segmentation network $\boldsymbol{G}$ to guide the class-relevant feature selection. Let $\boldsymbol{H}_s^k \in \mathbb{R}^{W\times H\times C}$ and $\boldsymbol{H}_t^k \in \mathbb{R}^{W\times H\times C}$ denote source and target domain features extracted from the kth intermediate hidden layer of network $\boldsymbol{G}$, respectively. Let $\tilde{y}_s^i \in \mathbb{R}^{W_0\times H_0\times C_0}$ and $\tilde{y}_t^j \in \mathbb{R}^{W_0\times H_0\times C_0}$ denote the network's predicted score maps of the source and target domain examples, respectively. W and H are the width and height of the feature map, respectively, and C is the number of channels of the feature map. W_0 and H_0 are the width and height of the predicted score map, respectively, and C_0 represents the class number. In this article, we simply set $C_0 = 2$, i.e., cloud and noncloud category. Therefore, cloud-relevant feature $\boldsymbol{F}_s^k \in \mathbb{R}^{W\times H\times C}$ and noncloud-relevant feature $\boldsymbol{B}_s^k \in \mathbb{R}^{W\times H\times C}$ of source domain feature $\boldsymbol{H}_s^k \in \mathbb{R}^{W\times H\times C}$ are obtained by

$$\boldsymbol{F}_s^k = \boldsymbol{H}_s^k \otimes \mathcal{F}\left(\tilde{y}_s^i(:,:,1)\right) \tag{1}$$

$$\boldsymbol{B}_s^k = \boldsymbol{H}_s^k \otimes \mathcal{F}\left(\tilde{y}_s^i(:,:,2)\right) \tag{2}$$

where $\tilde{y}_s^i(:,:,1)$ and $\tilde{y}_s^i(:,:,2)$ are the spatial attention maps of cloud and noncloud areas of source domain data, respectively. $\mathcal{F}(\cdot)$ represents the downsample or upsample operator that makes the predicted score map with the same resolution as hidden-layer feature $\boldsymbol{H}_s^k$. $\otimes$ represents the elementwise multiplication operation. During multiplication, the spatial attention values are broadcasted (copied) along the channel dimension.

Similarly, cloud-relevant feature $\boldsymbol{F}_t^k \in \mathbb{R}^{W\times H\times C}$ and noncloud-relevant feature $\boldsymbol{B}_t^k \in \mathbb{R}^{W\times H\times C}$ of target domain feature $\boldsymbol{H}_t^k \in \mathbb{R}^{W\times H\times C}$ are obtained by

$$\boldsymbol{F}_t^k = \boldsymbol{H}_t^k \otimes \mathcal{F}\left(\tilde{y}_t^i(:,:,1)\right) \tag{3}$$

$$\boldsymbol{B}_t^k = \boldsymbol{H}_t^k \otimes \mathcal{F}\left(\tilde{y}_t^i(:,:,2)\right) \tag{4}$$

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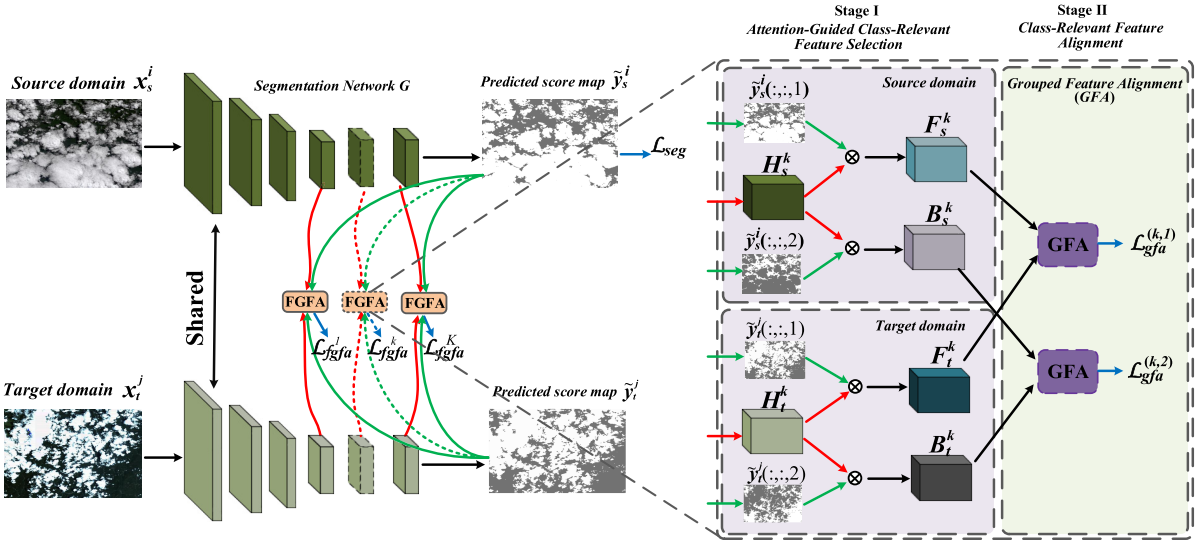

Fig. 3. Proposed UDA framework for cloud detection of RS images. In this framework, we propose an FGFA strategy to align the local class-relevant features in source and target domain datasets. The proposed FGFA domain adaptation consists of two stages: 1) AG class-relevant feature selection and 2) class-relevant FA. During the training process, segmentation network $\boldsymbol{G}$ of the proposed UDA framework has two input datasets, i.e., source domain dataset $X_s = \{x_s^i\}_{i=1}^{N^s}$ with corresponding label $Y_s = \{y_s^i\}_{i=1}^{N^s}$ and the target domain dataset $X_t = \{x_t^j\}_{j=1}^{N^t}$ without labels, where x_s^i and x_t^i are three channel images and N^s and N^t represent the number of samples in source and target domain datasets, respectively. $\tilde{y}_s^i$ and $\tilde{y}_t^i$ are the predicted score maps. $\tilde{y}_s^i(:,:,1)$ and $\tilde{y}_s^i(:,:,2)$ are the spatial attention maps of cloud and noncloud areas of source domain data, respectively. $\tilde{y}_t^i(:,:,1)$ and $\tilde{y}_t^i(:,:,2)$ are the spatial attention maps of cloud and noncloud areas of target domain data, respectively. $\boldsymbol{F}_s^k$ and $\boldsymbol{B}_s^k$ are cloud-relevant feature and noncloud-relevant feature of source domain feature $\boldsymbol{H}_s^k$, respectively. $\boldsymbol{F}_t^k$ and $\boldsymbol{B}_t^k$ are cloud-relevant feature and noncloud-relevant feature of target domain feature $\boldsymbol{H}_t^k$, respectively. The objective function of the kth hidden-layer feature is defined as $\mathcal{L}_{\text{fgfa}}^k = \mathcal{L}_{\text{gfa}}^{(k,1)} + \mathcal{L}_{\text{gfa}}^{(k,2)}$, where $\mathcal{L}_{\text{gfa}}^{(k,1)}$ and $\mathcal{L}_{\text{gfa}}^{(k,2)}$ represent GFA losses of the cloud-relevant feature and noncloud-relevant feature, respectively. The red arrows are used for hidden-layer feature and green arrows for predicted score maps. The blue arrows represent objective losses. The symbol ⊗ represents the elementwise multiplication operation. Best viewed in color.

where $\tilde{y}_t^i(:,:,1)$ and $\tilde{y}_t^i(:,:,2)$ are the spatial attention maps of cloud and noncloud areas of target domain data, respectively. After obtaining cloud and noncloud relevant features, we further apply the proposed GFA strategy to align these features effectively.

C. Class-Relevant FA

To achieve a promising FA performance, instead of traditional HFA, we propose a GFA strategy to effectively align the class-relevant features in the source and target domain datasets. As shown in Fig. 4, inspired by grouped convolution [23], we first split the obtained class-relevant feature into a series of groups and then apply FA by using adversarial learning strategy separately for each grouped feature in source and target domains. Actually, these operations perform local class-relevant FA rather than holistic class-relevant FA. Formally, given two cloud-relevant feature maps extracted from the kth hidden layer of the network $\boldsymbol{G}$, i.e., source domain feature map $\boldsymbol{F}_s^k \in \mathbb{R}^{W\times H\times C}$ and target domain feature map $\boldsymbol{F}_t^k \in \mathbb{R}^{W\times H\times C}$, we first conduct feature split operation to obtain the feature map groups

$$\boldsymbol{F}_s^k \xrightarrow{\text{split}} \left\{\boldsymbol{f}_s^{(k,1)}, \boldsymbol{f}_s^{(k,2)}, \ldots, \boldsymbol{f}_s^{(k,R)}\right\} \tag{5}$$

$$\boldsymbol{F}_t^k \xrightarrow{\text{split}} \left\{\boldsymbol{f}_t^{(k,1)}, \boldsymbol{f}_t^{(k,2)}, \ldots, \boldsymbol{f}_t^{(k,R)}\right\} \tag{6}$$

where $R \in 2^n$ represents the number of feature map groups. $\boldsymbol{f}_s^{(k,r)} \in \mathbb{R}^{W\times H\times (C/R)}$ and $\boldsymbol{f}_t^{(k,r)} \in \mathbb{R}^{W\times H\times (C/R)}$ represent the grouped features, which are further aligned by classical domain adversarial method [47].

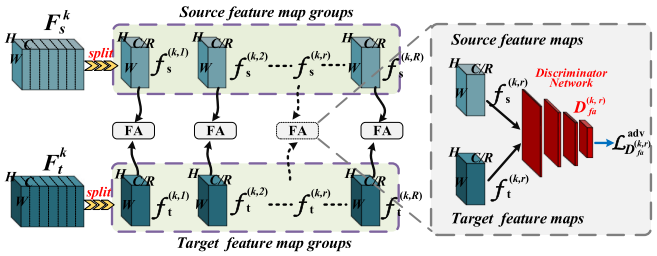

Fig. 4. Proposed GFA domain adaptation strategy. We split intermediate feature maps into a series of groups and apply FA domain adaptation using an adversarial learning strategy separately for each group in source and target data. The discriminator $\boldsymbol{D}_{\text{fa}}^{(k,r)}$ is colored with red. Best viewed in color.

In this article, we design a standard binary classification network as the discriminator $\boldsymbol{D}_{\text{fa}}^{(k,r)}$. It can help network $\boldsymbol{G}$ to generate domain-invariant feature representations. During the discriminator $\boldsymbol{D}_{\text{fa}}^{(k,r)}$ training, we use a cross-entropy domain classification loss [44] as the objective function. It can be defined as

$$\mathcal{L}_{D_{\text{fa}}^{(k,r)}} = -\mathbb{E}_{\boldsymbol{I}_x^s}\left[\log\left(D_{\text{fa}}^{(k,r)}\left(\boldsymbol{f}_s^{(k,r)}\right)\right)\right] - \mathbb{E}_{\boldsymbol{I}_x^t}\left[\log\left(1 - D_{\text{fa}}^{(k,r)}\left(\boldsymbol{f}_t^{(k,r)}\right)\right)\right]. \tag{7}$$

For network $\boldsymbol{G}$ training, the adversarial objective provided by discriminator $\boldsymbol{D}_{\text{fa}}^{(k,r)}$ is defined as follows:

$$\mathcal{L}_{D_{\text{fa}}^{(k,r)}}^{\text{adv}} = -\mathbb{E}\left[\log D_{\text{fa}}^{(k,r)}\left(\boldsymbol{f}_t^{(k,r)}\right)\right] \tag{8}$$

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where $\mathcal{L}^{\text{adv}}_{D^{(k,r)}_{\text{fa}}}$ represents the rth adversarial objective. Since the proposed GFA domain adaptation contains a series of sub-FA operations, the GFA loss is provided from all sub-FA losses. For GFA loss of the cloud-relevant feature, it is defined as

$$\mathcal{L}^{(k,1)}_{\text{gfa}} = \sum_{r=1}^{R} \mathcal{L}^{\text{adv}}_{D^{(k,r)}_{\text{fa}}}. \tag{9}$$

According to the above modeling process, we can easily obtain the GFA loss $\mathcal{L}^{(k,2)}_{\text{gfa}}$ of the noncloud-relevant feature.

D. Network Optimization Learning

During the training process, segmentation network $\boldsymbol{G}$ of the proposed UDA framework has two input datasets, i.e., source domain dataset $X_s = \{x^i_s\}^{N^s}_{i=1}$ with corresponding label $Y_s = \{y^i_s\}^{N^s}_{i=1}$ and the target domain dataset $X_t = \{x^j_t\}^{N^t}_{j=1}$ without labels, where N^s and N^t represent the number of samples in source and target domain datasets, respectively. The segmentation network $\boldsymbol{G}$ is supervised by a standard cross-entropy loss $\mathcal{L}_{\text{seg}}$ and a series of FGFA losses $\{\mathcal{L}^k_{\text{fgfa}}\}^K_{k=1}$

$$\mathcal{L}_G = \lambda_{\text{seg}}\mathcal{L}_{\text{seg}} + \lambda_{\text{fgfa}} \sum_{k=1}^{K} \mathcal{L}^k_{\text{fgfa}} \tag{10}$$

where λ_{seg} and λ_{fgfa} are regulation parameters used for minimizing the objective $\mathcal{L}_G$ and K represents the number of hidden-layer feature where FA domain adaptation is applied. $\mathcal{L}^k_{\text{fgfa}}$ represents the FGFA loss of the kth hidden-layer feature, which is defined as

$$\mathcal{L}^k_{\text{fgfa}} = \mathcal{L}^{(k,1)}_{\text{gfa}} + \mathcal{L}^{(k,2)}_{\text{gfa}} \tag{11}$$

where $\mathcal{L}^{(k,1)}_{\text{gfa}}$ and $\mathcal{L}^{(k,2)}_{\text{gfa}}$ are GFA losses of cloud-relevant feature and noncloud-relevant feature, respectively. In addition, the standard cross-entropy loss provided from labeled source example (x^s_i, y^i_s) is defined as follows:

$$\mathcal{L}_{\text{seg}} = -\sum_{h,w,c} y^i_s(w,h,c) \log \tilde{y}^i_s(w,h,c) \tag{12}$$

where $\tilde{y}^i_s = \boldsymbol{G}(x^i_s)$ is the predicated result.

In addition, the target samples are not only used to train segmentation network $\boldsymbol{G}$ but also used to minimize objectives of a series of class-relevant FA discriminators, such as the objectives of cloud-relevant FA discriminators $\{\mathcal{L}_{D^{(k,r)}_{\text{fma}}}\}^{K,R}_{k=1,r=1}$.

E. Network Structure

1) Segmentation Network: In this article, we use DeepLabv2 [29] as our main cloud detection segmentation network $\boldsymbol{G}$. In addition, we adopt ResNet-101 [25] as backbone architecture to extract features for cloud detection. At the last layer of backbone, there is an atrous spatial pyramid pooling (ASPP) module [29] that is designed to act as a multiscale feature extractor as well as a final classifier. In this network, the directly predicted score map is $1/8 \times 1.8$ size of input image. The final result is upsampled to match the size of the input image.

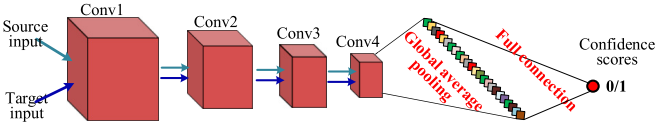

Fig. 5. Structure of the proposed discriminator network for domain adversarial learning.

2) Discriminator: The discriminative network is designed with a standard binary classifier. Fig. 5 shows the detailed structure of the proposed discriminator. It contains four convolutional layers with as 4×4 kernels, stride sets 2, and {256, 192, 128, 64} channels. Besides, after each convolutional layer, there is an activated layer. In this article, we use Leaky-ReLU activation (slope = 0.2). To avoid overfitting and improve training stability, we introduce a dropout layer (dropout rate = 0.5) for each convolutional layer. At the end of convolution operations, a global average pooling layer and a fully connected layer are introduced to obtain confidence scores for each input feature from different domains.

IV. Dataset and Experimental Settings

A. Experimental Dataset

To evaluate the proposed UDA method, we adapt two large-volume RS cloud detection datasets, Landsat-8 Cloud Cover Assessment Validation dataset[1] and GF-1 WFV Cloud and Cloud Shadow Cover Validation dataset,[2] as the source domain dataset, while the ZY-3 Satellite Thumbnail dataset as target domain dataset.

1) Target Domain Dataset: In this article, we collect 300 scene ZY-3 thumbnails, including 250 unlabeled thumbnails for network training and 50 labeled ones for model evaluation. Land-cover types of these training and testing images include forest, barren, water, snow/ice, urban, wetlands, and so on, as shown in Fig. 6. All 300 scenes ZY-3 thumbnails were collected from website of LSRSAC, Ministry of Natural Resources of P. R. China.[3] These 50 labels are manually labeled by personnel trained by experts from LSRSAC. Since ZY-3 satellite carries four cameras with different resolutions, the thumbnails of original RS images also show different image resolutions. Besides, the number of bands of obtained thumbnails is different due to the spectrum perception of different cameras. Therefore, there are large intradomain discrepancy gaps within the target ZY-3 thumbnail dataset. This is the reason for choosing feature-level domain adaptation instead of pixel-level (visual appearance) domain adaptation. In Table I, we present the detailed information of the ZY-3 thumbnail dataset. During training, we conduct data augmentation, such as image rotation, flipping, and cross cropping [10], to improve the robustness of the network. After data augmentation, there are about 19k unlabeled subimages of size 321×321 in the ZY-3 thumbnail dataset. It should be noted that single-channel gray thumbnails are broadcasted (copied) along the

[1] https://landsat.usgs.gov/landsat-8-cloud-cover-assessment-validation-data
[2] http://sendimage.whu.edu.cn/en/mfc-validation-data/
[3] http://sasclouds.com/english/normal/

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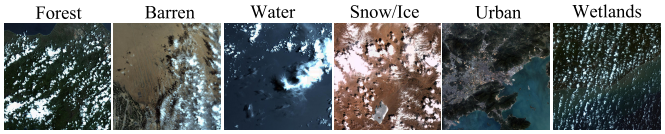

Fig. 6. Different land-cover types of ZY-3 thumbnail examples.

TABLE I
DETAILED INFORMATION OF ZY-3 THUMBNAILS

Image type	Image size	Spatial Resolution (m)
Nadir (Pan)	3.1k×3.1k	16
For-/Backward (Pan)	2.1k×2.1k	25
RGB (Multispectral)	1.2k×1.2k	48

[1] The spatial resolution of thumbnail is roughly estimated from the original image. The detailed information of ZY-3 satellite image is available at http://sasclouds.com/english/satellite/chinese/zy3.

channel dimension to generate three channels images for model training and evaluation.

2) Source Domain Datasets: The original Landsat-8 OLI and GF-1 WFV dataset contain 96 and 108 scenes multispectral images, respectively. Landsat-8 OLI multispectral image contains nine bands information and has a spatial resolution of 30 m (except Panchromatic band 8) [14]. GF-1 satellite carries a wide field of view (WFV) imaging system onboard with acquired GF-1 WFV images containing blue, green, red, and NIR bands and having a spatial resolution of 16 m [15]. Since the number of subimages with and without cloud in the training data should be balanced and the examples incorrectly annotated should be removed and thus cannot be used as training data, similar to [10], we extract only a subset out of these datasets for experiments. To be specific, we select 40 and 60 scenes ideal images (cloud coverage percentage: 5%–100%) from the Landsat-8 OLI and GF-1 WFV datasets, respectively. All training data are cropped into subimages of size 321 × 321. Finally, there are about 25k and 35k annotated subimages in the Landsat-8 OLI and GF-1 WFV training dataset, respectively.

In summary, the detailed information of these three datasets is listed in Table II. In this article, we have two domain-adaptation setups, i.e., "Landsat-8 → ZY-3" and "GF-1 → ZY-3." We simply use band 2 (blue), band 3 (green), and band 4 (red) of Landsat-8 OLI data and band 1 (blue), band 2 (green), and band 3 (red) of Gaofen-1 WFV data for domain adaptation tasks by considering the limited number of channels of ZY-3 thumbnail.

B. Network Training Details

Our proposed UDA method is developed based on the Pytorch framework. It is trained under the operating system Ubuntu 14.04 equipped with a NVIDIA GTX 1080 Ti GPU. We use the SGD [57] algorithm to optimize the generator network. In the two domain adaptation tasks, we set learning_rate $= 2.5 \times 10^{-4}$ for generator network training and learning_rate $= 1.0 \times 10^{-4}$ for discriminator network training. Meanwhile, we set training_decay = "poly" [58], batch_size = 4, momentum = 0.9, and weight_decay $= 5 \times 10^{-4}$. In this article, all models are fine-tuned with the model pretrained on the ImageNet dataset [59].

TABLE II
DETAILED INFORMATION OF THESE THREE DATASETS

Domain Type	Dataset	Dataset Scale	Data Size	Spatial Resolution
Source	GF-1	35k	321×321	16 m
Source	Landsat-8	25k	321×321	30 m
Target	ZY-3	19k	321×321	16/25/48 m

[1] In this paper, land-cover types of these three datasets include forest, barren, water, snow/ice, urban, wetlands and so on.

During training, we empirically set the number of feature groups $R = 4$ for all tasks and set the balance weights $\lambda_{\text{fgfa}} = 0.001$ and $\lambda_{\text{seg}} = 0.5$ for "Landsat-8 → ZY-3" domain adaptation task and $\lambda_{\text{fgfa}} = 0.0005$ and $\lambda_{\text{seg}} = 0.5$ for "GF-1 → ZY-3" domain adaptation task. We set $K = 2$, i.e., we apply FA domain adaptation at two hidden-layers' features. To be specific, we apply FGFA domain adaptation at the end feature maps of Conv4_x and Conv5_x layers of RestNet-101 for all domain adaptation tasks. The proposed UDA method is trained for 60k and 40k iterations on "Landsat-8 → ZY-3" and "GF-1 → ZY-3" domain adaptation tasks, respectively.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Ablation Study

To investigate the performance of the proposed UDA framework, we conduct a series of ablation studies to demonstrate the effects of each proposed component. All the experiments are conducted under the domain adaptation task of "Landsat-8 → ZY-3." Similar to [10] and [13], we use mean intersection over union (MIoU), kappa coefficient (Kappa), overall accuracy (OA), producer accuracy (PA), and user accuracy (UA) to quantitatively evaluate the proposed UDA method.

1) Ablation Study on Different Intermediate Layers: In this article, we focus on feature-level domain adaptation to reduce domain gaps between the source and target domains. Hence, it is important to explore which intermediate layers' feature should be aligned. We conduct experiments using class-relevant features that are obtained by AG class-relevant feature selection mechanism. In Table III, we list the experiment results on different intermediate layers. It is worth noting that results are obtained by using a traditional HFA strategy.

From Table III, we find that feature-level domain adaptation at the end of Conv4_x and Conv5_x residual blocks, i.e., AG_45+HFA, shows the best performance. In contrast, the feature domain adaptation applied at other layers' combinations always shows some suboptimal results. Therefore, we apply FA domain adaptation at the end of Conv4_x and Conv5_x residual blocks to conduct the following domain adaptation tasks.

2) Ablation Study on Both AG and GFA: To demonstrate the effectiveness of the AG mechanistic, we list the results of experiment conducted without using AG mechanistic,

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TABLE III

ABLATION STUDY OF DOMAIN ADAPTATION ON DIFFERENT INTERMEDIATE LAYERS

Methods	OA	MIoU	Kappa	PA	UA
AG_5+HFA	86.77	73.70	63.99	72.27	84.79
AG_45+HFA	**89.94**	**79.72**	**72.12**	**83.03**	82.91
AG_345+HFA	87.74	75.58	66.34	78.09	80.65
AG_2345+HFA	89.73	78.83	70.49	76.72	**87.12**

[1] We simply denote AG, which represents an abbreviation of $attention\ guided$, as attention guided (AG) class-relevant feature alignment domain adaptation without using grouped feature alignment strategy.

TABLE IV

ABLATION STUDY ON AG MECHANISTIC

Methods	OA	MIoU	Kappa	PA	UA
without_AG_45+HFA	86.71	73.48	63.23	**85.89**	74.49
AG_45+HFA	89.94	79.72	72.12	83.03	82.91
without_AG_45 + GFA	89.69	79.59	70.23	85.87	80.67
AG_45 + GFA	**91.27**	**82.16**	**75.21**	85.38	**85.30**

as shown in Table IV. The experiment results indicate that AG mechanistic ("AG_45+HFA") helps achieve better performance than without AG mechanistic ("without_AG_45+HFA"). The primary reason for this large performance gap is that "without_AG_45+HFA" focuses on aligning the global feature marginal distribution between source and target domains while ignoring the domain distribution shift problems of local categorywise feature. In this article, the proposed AG mechanistic plays an important role in suppressing class-irrelevant feature and activating (selecting) class-relevant feature for applying FA.

To demonstrate the effectiveness of the GFA, we also report the cloud detection performance of the proposed GFA method in Table IV. Results show that the proposed GFA brings significant performance improvement on both "without_AG_45+GFA" and "AG_45+GFA" tasks. We find that the proposed GFA obtains the improvement of $+2.98\%$, $+6.11\%$, and $+7.0\%$ in OA, MIoU, and Kappa, respectively, under the global feature marginal distribution alignment ("without_AG_45+GFA"). Meanwhile, GFA obtains the improvement of $+1.33\%$, $+2.44\%$, and $+3.09\%$ in OA, MIoU, and Kappa, respectively, under the local class-relevant feature distribution alignment ("AG_45+GFA"). Results in Table IV demonstrate that the FA ability of GFA is superior to that of traditional HFA method. In addition, from Table IV, we can further find that combining class-relevant feature selection method with GFA domain adaptation strategy, i.e., AG_45+GFA, enables model trained on a source dataset to be well generalized to target dataset. Our proposed "AG_45+GFA" has increased Kappa from 70.23% achieved with "without_AG_45+GFA" to 75.21%. Results in Table IV have demonstrated that the proposed "AG_45+GFA" is a promising framework to realize an FGFA. In summary, the proposed GFA plays a key role in aligning class-relevant feature between the source and target domain datasets.

TABLE V

ABLATION STUDY ON THE NUMBER (R) OF FEATURE MAP GROUPS

Groups	OA	MIoU	Kappa	PA	UA
$R = 1$	89.94	79.72	72.12	83.03	82.91
$R = 2$	89.83	79.69	71.15	83.21	82.05
$R = 4$	**91.27**	**82.16**	**75.21**	85.38	**85.30**
$R = 8$	90.69	81.49	74.01	**88.34**	80.31
$R = 16$	89.79	78.87	71.29	82.94	81.41

TABLE VI

STUDY ON HYPERPARAMETER λ_{seg} AND λ_{fgfa}

λ_{seg}	λ_{fgfa}	OA	MIoU	Kappa	PA	UA
0.6	0.0014	89.85	81.66	73.80	85.68	83.51
0.6	0.0012	90.12	82.02	74.39	85.27	84.24
0.6	0.0010	91.02	82.10	74.88	85.20	84.69
0.6	0.0008	90.66	82.05	74.56	84.80	84.67
0.6	0.0006	89.79	81.46	73.73	85.04	84.20
0.5	0.0014	90.17	81.61	74.63	84.80	85.31
0.5	0.0012	91.19	82.14	75.15	85.21	85.14
0.5	0.0010	**91.27**	**82.16**	**75.21**	85.38	**85.30**
0.5	0.0008	91.22	82.12	75.17	85.20	85.22
0.5	0.0006	90.14	81.30	74.13	85.07	85.18
0.4	0.0014	89.87	81.80	74.08	84.62	84.96
0.4	0.0012	91.01	82.10	75.08	84.90	85.29
0.4	0.0010	91.11	82.12	75.15	**85.40**	84.79
0.4	0.0008	90.29	81.70	74.74	84.69	85.10
0.4	0.0006	89.79	81.61	74.02	84.21	85.24

3) Ablation on the Number (R) of Feature Map Groups: To conduct GFA domain adaptation, we split feature maps into a series of subfeature maps along the channel axis. Hence, the number (R) of the feature map groups is an important parameter that affects the effectiveness of GFA. In Table V, we present the results of different R. Results show that $R = 4$ achieves the best performance. Whether the group quantity is overmuch or insufficient, it can lead to unsatisfactory cloud detection results. The reason is that overmuch quantity of groups (subdomains) causes the model to fall into specific local optimum and ignore global optimum, thus reducing the generalization ability of the model. Meanwhile, insufficient quantity of groups makes model prone to under transfer when aligning class-relevant feature between the source and target domain datasets.

4) Impact of Balance Weights: The training objective $\mathcal{L}_G$ has two balance weights, i.e., λ_{seg} and λ_{fgfa}. We conduct a series of validation experiments to investigate the impact of these two weights on our UDA framework. Table VI lists partial results. From Table VI, we can find that the promising cloud detection performance is achieved by the setting of $\lambda_{\text{seg}} = 0.5$ and $\lambda_{\text{fgfa}} = 0.001$. It can be seen that no matter what we fix the weight of $\lambda_{\text{seg}} = 0.5$ and raise/reduce the value of λ_{fgfa} or fix the weight of $\lambda_{\text{fgfa}} = 0.001$ and raise/reduce the value of λ_{seg}, results all show decreased performance. Finally, we empirically set the balance weights $\lambda_{\text{seg}} = 0.5$ and $\lambda_{\text{fgfa}} = 0.001$ for "Landsat-8 $\rightarrow$ ZY-3" task.

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TABLE VII
QUANTITATIVE COMPARISON RESULTS OF LANDSAT-8 → ZY-3 DOMAIN ADAPTATION TASK

Methods	OA	MIoU	Kappa	PA	UA
Source Model [29]	83.99	67.30	48.75	52.06	**87.82**
ADDA [18]	85.85	73.48	63.33	**85.88**	74.49
AdvEnt [20]	89.96	80.03	70.92	83.64	83.77
FADA [22]	90.12	80.32	71.69	83.72	83.82
Our method	**91.27**	**82.16**	**75.21**	85.38	85.30

TABLE VIII
QUANTITATIVE COMPARISON RESULTS OF GF-1 → ZY-3 DOMAIN ADAPTATION TASK

Methods	OA	MIoU	Kappa	PA	UA
Source Model [29]	86.16	70.70	53.55	52.03	95.98
ADDA [18]	88.95	76.75	65.05	64.41	**95.26**
AdvEnt [20]	90.28	81.05	71.60	74.81	91.19
FADA [22]	91.44	82.03	72.38	74.78	90.47
Our method	**92.01**	**83.09**	**75.29**	**79.85**	87.53

B. Comparisons With State-of-the-Art Methods

For comprehensive evaluation, we compare with a feature-level domain adaptation approach, i.e., ADDA method [18], an output-level domain adaptation approach, i.e., AdvEnt minimization domain adaptation [20], and a class-level FA method based on fine-grained adversarial learning strategy, named FADA [22]. For a fair comparison, we use the DeepLabv2 [29] as the baseline for all UDA methods. All the competing methods are trained under their optimal settings to achieve their promising cloud detection performance on the target domain dataset.

1) Landsat-8 → ZY-3: In Table VII, we list cloud detection results for different methods in "Landsat-8 → ZY-3" domain adaptation task. Results in Table VII show that our FGFA domain adaptation method outperforms the competing methods in terms of OA, MIoU, and Kappa. Our proposed method has an obvious advantage in terms of Kappa (75.21%) and also has the best OA (91.27%) and MIoU (82.16%) accuracy among all competing methods. To be specific, our method outperforms the second-best FADA [22] method by 1.15%, 1.84%, and 3.52% in terms of OA, MIoU, and Kappa, respectively. Compared with ADDA [18] and AdvEnt [20], our proposed method has obvious gains in terms of OA, MIoU, Kappa, and UA. In addition, our method obtains significant improvement of +7.28%, +14.86%, +26.46%, and +32.32% in OA, MIoU, Kappa, and PA, respectively, compared with the source model. These results indicate that the proposed FGFA domain adaptation approach is an effective FA method in the "Landsat-8 → ZY-3" domain adaptation task.

In Fig. 7, we present the qualitative results of different methods. It can be seen that the predicted cloud masks from the source model suffer from severe misclassification except for Fig. 7(g). After applying domain adaptations, i.e., ADDA [18], AdvEnt [20], FADA [22], and our FGFA domain adaptation methods, predicted results are significantly improved. These results indicate that applying domain adaptation is helpful to improve the source-trained model's generalization performance. Meanwhile, it can be found from the results that there are fewer misclassified pixels with our method being used, indicating that our method has the best performance in reducing the feature domain gaps between the source and target domain datasets.

2) GF-1 → ZY-3: In Table VIII, we also report the cloud detection results for different methods in "GF-1 → ZY-3" domain adaptation task. From Table VIII, we find that the proposed method still achieves outstanding cloud detection performance. Our method achieves the best results in terms of OA, MIoU, Kappa, and PA. We obtain 83.09% mIoU that is 12.39% higher than the one obtained by the source model without adaptation. Compared with ADDA [18], our method also has obvious advantages in terms of OA, MIoU, Kappa, and PA, which indicates the effectiveness of our FGFA domain adaptation method. In terms of OA, MIoU, Kappa, and PA, our method outperforms the ADDA [18] by 3.06%, 6.34%, 10.24%, and 15.44%. In addition, our method also outperforms AdvEnt [20] and FADA [22].

In Table VIII, we find that PA values of all methods are lower than those of UA values, especially in the results of source model and ADDA [18], which indicates that there are many underdetected (omission) cloud pixels. To verify this problem, we visualize the detected results on their input images, as shown in Fig. 8. Experimental results show that many cloud pixels in these test images [except Fig. 8(e)] are underdetected, especially the pixels at the boundary of the cloud objects and the thin cloud areas [see Fig. 8(b)]. Nevertheless, results show that the detected cloud masks of our method show promising consistency with the cloud cover areas. There are fewer misclassified cloud pixels in our results than those achieved by competing methods, which can be found from the zoomed results of green rectangles. In summary, results in Table VIII and Fig. 8 have demonstrated the effectiveness of the proposed FGFA domain adaptation method.

C. Results Analysis and Discussion

1) Cloud Cover Estimation Accuracy: Cloudage (i.e., cloud cover percentage) of RS image is an important image quality index and is also taken as important retrieve information provided by data producers for users to effectively retrieve high-quality RS images from a huge RS database. To investigate the cloudage estimation performance of the proposed method, we present the cloudage estimation results on these evaluation thumbnails, as shown in Figs. 9 and 10. We use a series of discrete points to present the normalized estimated cloudages against the ground truth. More specifically, each discrete point is plotted by detected cloudage on the horizontal axis and ground truth (GT) cloudage on the vertical axis. We use a simple straight line to fit the discrete points. Mathematically, when cloudage estimation results are perfectly consistent with the GT, the plotted line has a slope of one (named benchmark line). From the results in Figs. 9 and 10,

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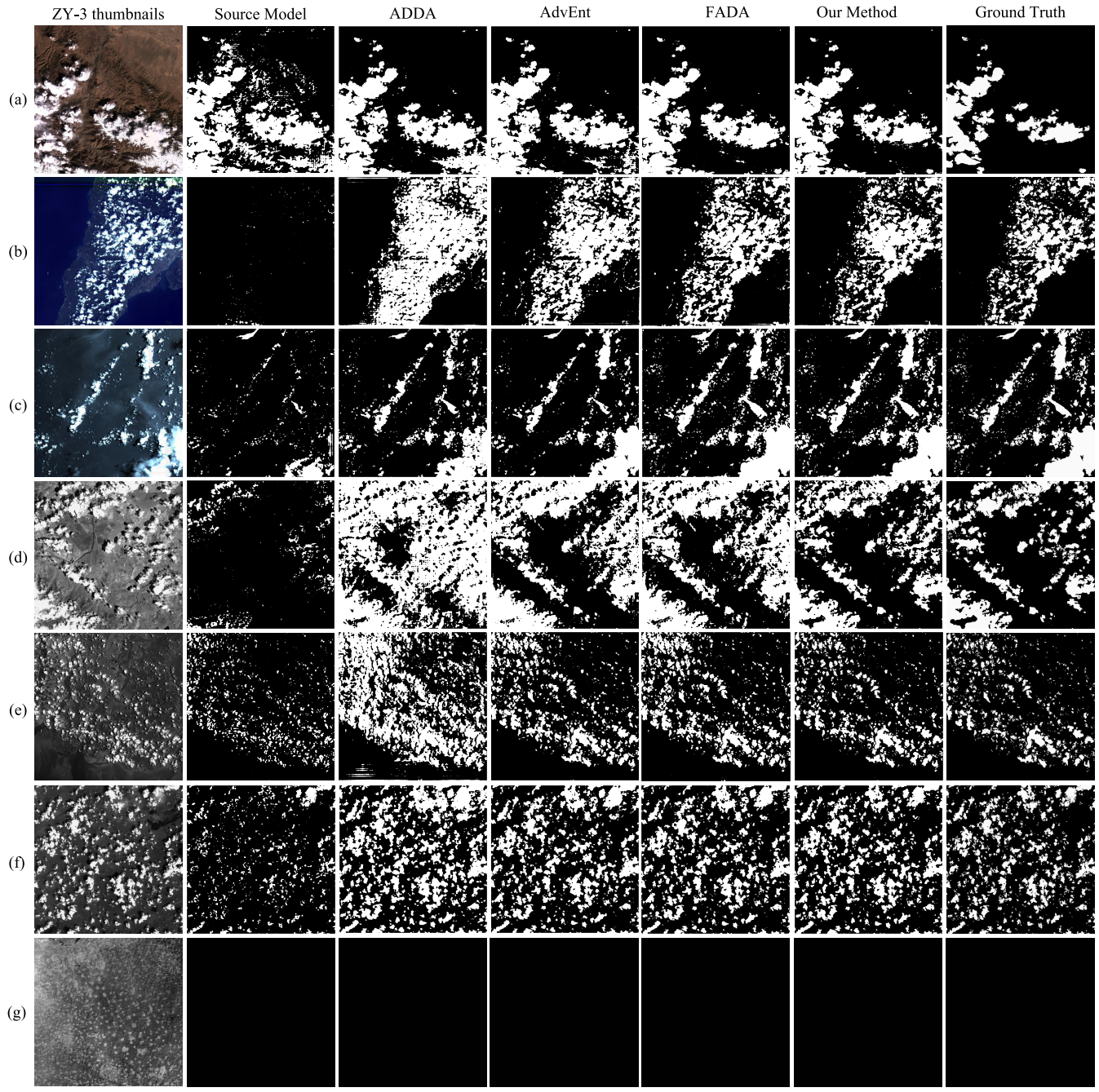

Fig. 7. Qualitative comparison results in "Landsat-8 → ZY-3" domain adaptation task. Seven ZY-3 thumbnail examples contain different land cover types, including (a) barren, snow, and mountain areas, (b) city, water, and forest areas, (c) water/ocean areas, (d) mountain, village, and wetlands areas, (e) water, wetlands, and village areas, (f) plains, city, and water areas, and (g) farmland, village, and city, where (a)–(c) are RGB images, while (d)–(g) are gray images. Zoomed-in view for better view.

we find that the fit lines achieved by the proposed method distribute more closer to the benchmark line than those achieved by competing methods. In addition, as an important index of curve fitting accuracy, R-square shows that our method has the best cloudage estimation performance. To be specific, compared with the second-best FADA [22], our method obtains 0.0576 and 0.0225 improvement, respectively, in terms of R-square for "Landsat-8 → ZY-3" and "GF-1→ ZY-3" domain adaptation tasks. Meanwhile, compared with ADDA [18], AdvEnt [20], and source model, our method has a significant improvement in terms of R-square for these two domain adaptation tasks. All of the results have demonstrated the effectiveness of the proposed UDA method.

2) Feature Representation Visualization: In Fig. 11, we visualize the feature representation results of a sample image by using the t-SNE [60] strategy in the "Landsat-8 → ZY-3" domain adaptation task. It can be seen that cloud and noncloud example feature representations are embedded into each group and the source/target representations overlap well [Fig. 11(b)] after applying domain adaptation. Results in

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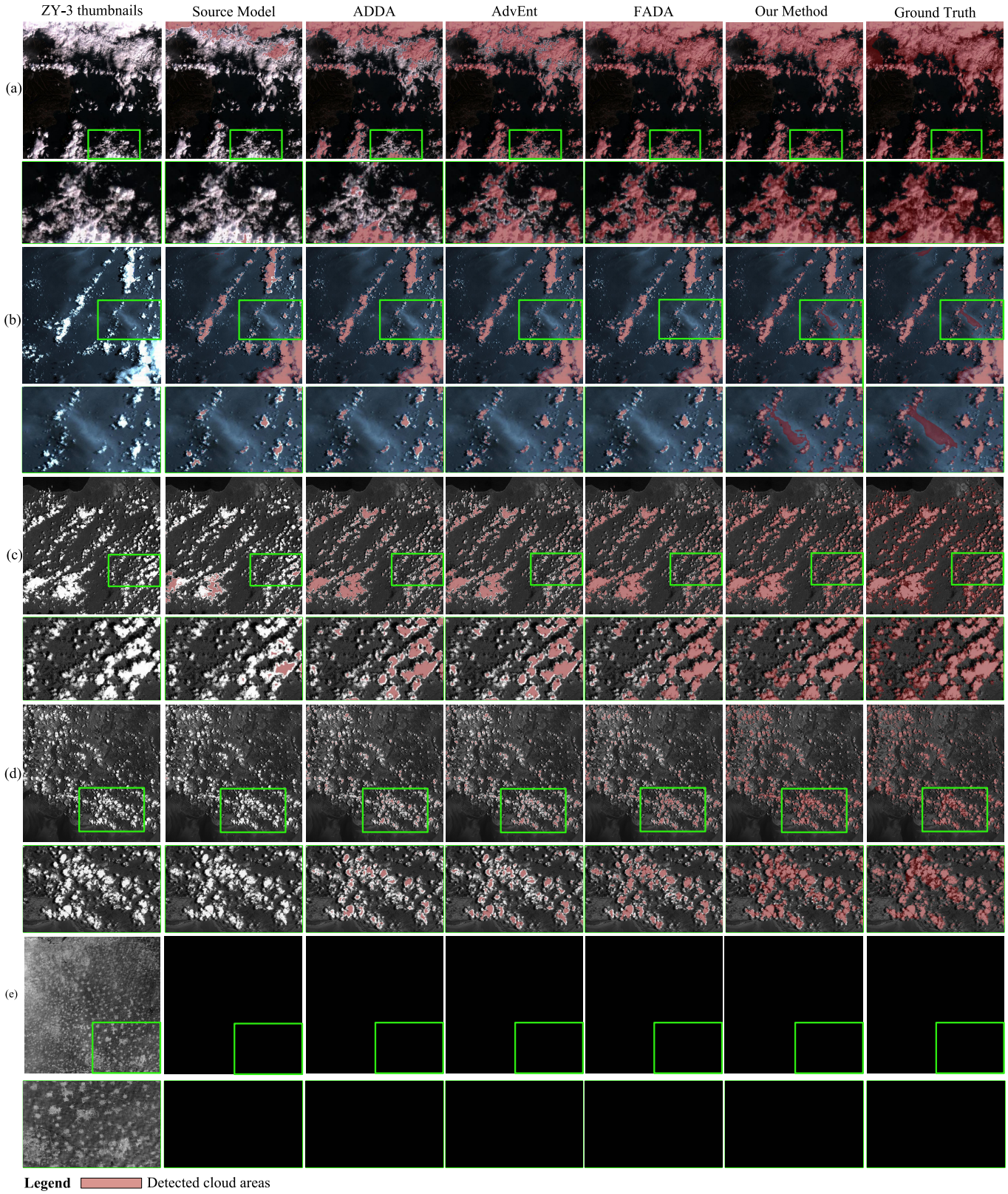

Fig. 8. Qualitative comparison results in "GF-1 → ZY-3" domain adaptation task. Five ZY-3 thumbnail examples contain different cloud shapes and categories, including (a) stratocumulus cloud (thick) with variable textures and shapes, (b) cirrus cloud (thin) and cumulus/cirrostratus cloud (thick), (c) altocumulus cloud (thick), (d) cirrocumulus cloud (thick), and (e) noncloud, where (a) and (b) are RGB images, while (c)–(e) are gray images. Each subimage in the green rectangular area is scaled up and displayed below the original image. Best viewed in colors.

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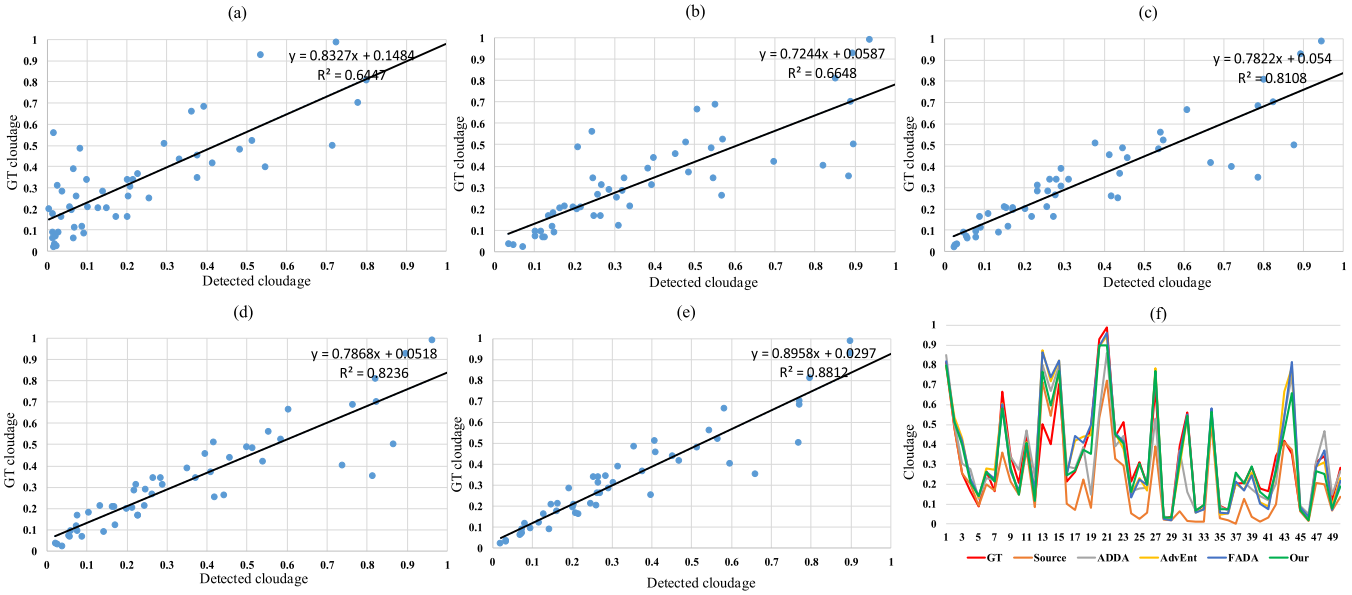

Fig. 9. Comparison of cloud cover estimation accuracy between (a) source model, (b) ADDA, (c) AdvEnt, (d) FADA, and (e) our method in "Landsat-8 → ZY-3" domain adaptation task. (f) Comparison of all results (detected cloudage of all methods).

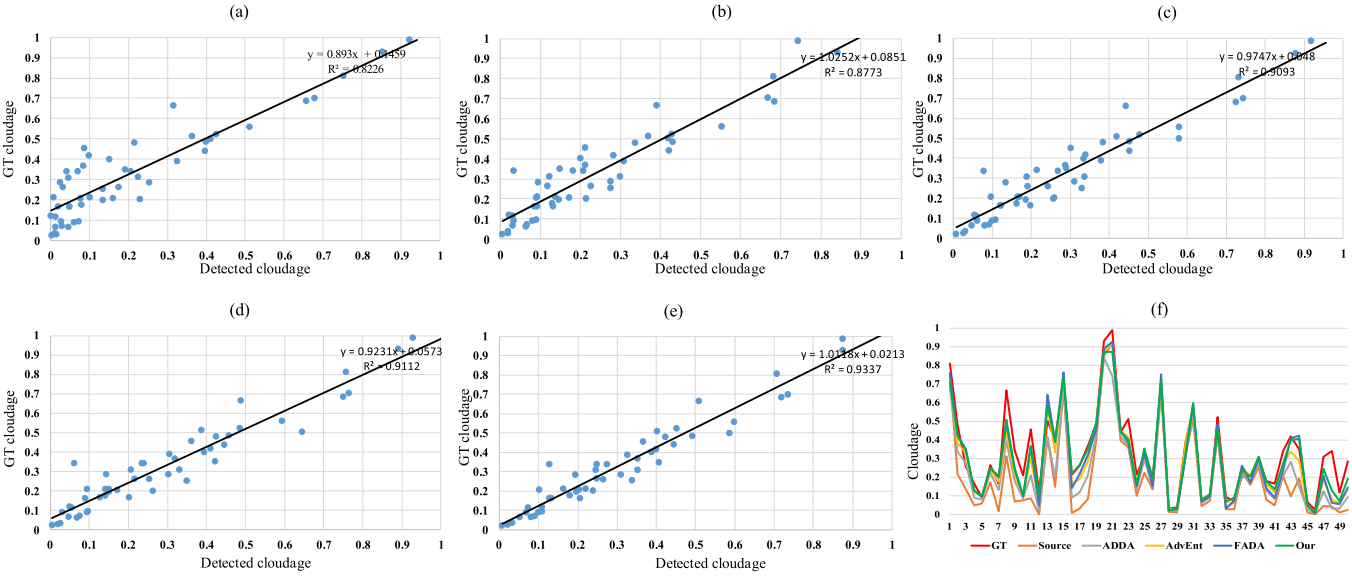

Fig. 10. Comparison of cloud cover estimation accuracy between (a) source model, (b) ADDA, (c) AdvEnt, (d) FADA, and (e) our method in "GF-1 → ZY-3" domain adaptation task. (f) Comparison of all results (detected cloudage of all methods).

Fig. 11(b) show that cloud and noncloud example features are more easier to discriminate. In contrast, before applying the proposed UDA method, the source/target feature representations are low in clustering and less overlapped as visualized in Fig. 11(a), which indicates that cloud and noncloud example features are difficult to distinguish. Experiment results in Fig. 11 demonstrate that applying domain adaptation is able to assign the same object features to the same cluster and improve segmentation accuracy.

D. Experimental Results on Landsat-8 and GF-1 Cloud Cover Assessment Validation Data

1) Domain Adaptation on "Landsat-8 ⇆ GF-1" Tasks: In this article, we also evaluate our proposed UDA method under the domain adaptation tasks of "Landsat-8 → GF-1" and "GF-1 → Landsat-8." For "Landsat-8 → GF-1" task, we use the 25k annotated Landsat-8 OLI subimages as the source dataset and 35k GF-1 WFV subimages as the target dataset. For "GF-1 → Landsat-8"task, we use the 35k annotated GF-1 WFV subimages as the source dataset and 25k Landsat-8 OLI subimages as the target dataset. In addition, we select 20 scenes Landsat-8 OLI and GF-1 WFV multispectral images for testing. In Table IX, we show the experiment results on these two domain adaptation tasks. Experiment results show that our method is still able to achieve promising detection performance and outperform the competing UDA methods.

2) Compared With Supervised Learning Methods: To further demonstrate the effectiveness of the proposed UDA method, we also compare our proposed UDA method with

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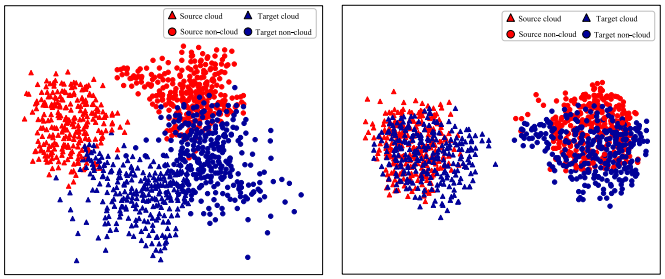

Fig. 11. Visualization of embedded feature representations. Note that (a) is the feature representations without domain adaptation and (b) is the feature representations with domain adaptation. Here, the end feature maps of Conv5_*x* layer of RestNet-101 are sampled as the example feature representations.

TABLE IX
QUANTITATIVE COMPARISON RESULTS OF "LANDSAT-8 ⇆ GF-1" DOMAIN ADAPTATION TASKS

Tasks	Methods	OA	MIoU	Kappa	PA	UA
Landsat-8→GF-1	Source Model [29]	88.41	66.16	51.91	44.63	**92.85**
	ADDA [18]	93.01	87.11	82.11	83.18	88.87
	AdvEnt [20]	93.40	87.51	83.15	85.48	90.82
	FADA [22]	94.11	88.12	84.28	86.39	90.30
	Our method	**94.80**	**88.91**	**86.01**	**87.42**	89.99
GF-1→Landsat-8	Source Model [29]	89.20	82.36	78.75	88.38	84.47
	ADDA [18]	92.43	86.50	83.69	89.92	92.31
	AdvEnt [20]	93.13	87.92	84.38	**91.15**	92.27
	FADA [22]	94.38	88.69	84.91	91.08	92.90
	Our method	**95.43**	**89.26**	**86.58**	90.12	**93.82**

TABLE X
QUANTITATIVE COMPARISON RESULTS OF SUPERVISED LEARNING TASKS IN LANDSAT-8 AND GF-1 DATASETS

Tasks	Methods	OA	MIoU	Kappa	PA	UA
GF-1	Our UDA method	94.80	88.91	86.01	87.42	89.99
	DeepLabV2 [29]	96.11	90.15	87.64	87.28	**92.15**
	CDnet [10]	**97.02**	**90.90**	**88.42**	**89.47**	91.66
Landsat-8	Our UDA method	95.43	89.26	86.58	90.12	**93.82**
	DeepLabV2 [29]	96.56	91.03	89.25	91.10	91.26
	CDnet [10]	**97.27**	**91.70**	**89.86**	**92.08**	91.41

two supervised learning methods (DeepLabV2 [29] and CDnet [10]). All the networks use ResNet-101 [25] as backbone architecture to extract features. We use the collected source datasets, i.e., 25k annotated Landsat-8 OLI subimages and 35k annotated GF-1 WFV subimages, as the training data. In Table X, we show the quantitative comparison results. Experimental results show that the accuracy of the proposed UDA method is close to but lower than those of the supervised learning methods. Therefore, for UDA methods, there is a long way to improve their performance. In future, we will focus on this area to further improve UDA approaches performance.

VI. CONCLUSION

In this article, we propose an FGFA domain adaptation method, which focuses on encouraging the segmentation network to learn domain-invariant feature from different domain datasets, thus encouraging the network trained on the source domain dataset to be well generalized to the target domain dataset. To be specific, we propose an AG class-relevant feature selection mechanism to select fine-grained class-relevant features in source and target domain datasets and propose the GFA strategy to effectively minimize class-relevant feature domain distribution gaps between different domains. We conduct domain adaptation experiments on "Landsat-8 → ZY-3," "GF-1→ ZY-3," and "Landsat-8 ⇆ GF-1" tasks. Experimental results show that the proposed FGFA domain adaptation method is able to achieve the state-of-the-art performance. This method provides a promising way for unsupervised cloud detection of RS images.

In the future, we will try to further improve the model's performance and explore UDA approaches for other object detection tasks, such as water, building, plant, and road. Besides, we will try to explore lightweight networks to reduce the running time and access memory of UDA model and focus on exploring source-free domain adaptation approach to avoid using any labeled dataset. Also, we will further explore the working mechanism of the UDA model.

ACKNOWLEDGMENT

The authors would like to thank Prof. Huanfeng Shen's team from Wuhan University, Wuhan, China, for providing the GF-1 WFV Cloud Cover Validation dataset. They would also like to thank the editors and reviewers for their valuable suggestions.

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